



Fermentation and Distillation

Distillation is the process of carefully evaporating the water and alcohol components of a fermented alcoholic beverage in such a way that an alcohol-rich and water-rich fraction can be separated. Repeated evaporations can produce a nearly pure (~95%) ethanol solution, which is then watered down and/or aged to produce distilled spirits such as brandy, vodka, gin, and whiskey; these are generally bottled at 40–50% alcohol. Both beers and wines can be distilled. For example, distilled wines are generally known as brandies (including both grape wine and other fruit wines, such as apricot brandy), while distilled grain beers are used to produce many vodkas and gins, as well as whiskey.

The history of distillation is much more debated than the history of wines and beers. A notable example comes from Mexico, where agave is cultivated for two alcoholic beverages: pulque, a low-alcohol beverage made from fermented agave (also termed "maguey") juice, and mezcal (of which one type is tequila), a distilled alcohol made from fermented agave juice that is then distilled.

Agave was cultivated for multiple reasons, for both the juice (whether fermented or drunk immediately as a sweet juice) and the fibers of the leaves. Learn more about how distillation happens and the debated history of agave distillation before the arrival of the Spanish to Mexico in the 50-minute movie below. Feel free to make yourself a [Oaxaca Old Fashioned](https://cooking.nytimes.com/recipes/1012967-oaxaca-old-fashioned) (<https://cooking.nytimes.com/recipes/1012967-oaxaca-old-fashioned>) to enjoy as you watch the film! As you watch the movie, take notes on three topics:

1. How does the distribution of different species of agave today inform our understanding of agave cultivation before the arrival of the Spaniards?
2. What is the traditional process by which mezcal is made in Mexico?
3. What is the evidence cited by these scholars that supports the independent development of distillation in Mexico prior to Spanish arrival?